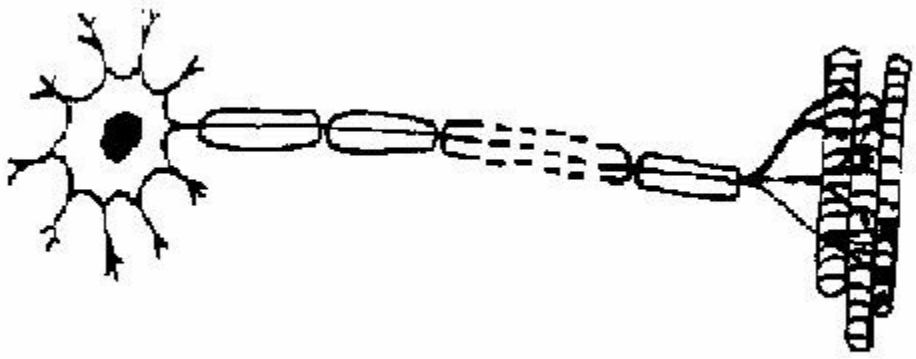
Neurons

1. The diagram below is of a nerve cell or neurone.

i. Add the following labels to the diagram.
Axon; Myelin sheath; Cell body; Dendrites; Muscle fibres;
ii. If you like, colour in the diagram as suggested below.
Axon - purple;
Myelin sheath - yellow;
Cell body - blue;
Dendrites - green;
Muscle fibres – red;
iii. Now indicate the direction that the nerve impulse travels.



2. There are three different kinds of neuron or nerve cell. Match each kind with its function.

A. Motor neuron; B. Sensory neuron; C. Relay neuron;

Kind of neuron	Function
.....	The nerve cell that carries impulses from a sense receptor to the brain or spinal cord.

.....	The nerve cell that connects sensory and motor neurons
.....	The nerve cell that transmits impulses from the brain or spinal cord to a muscle or gland

3. Match the descriptions in the table below with the terms in the list.

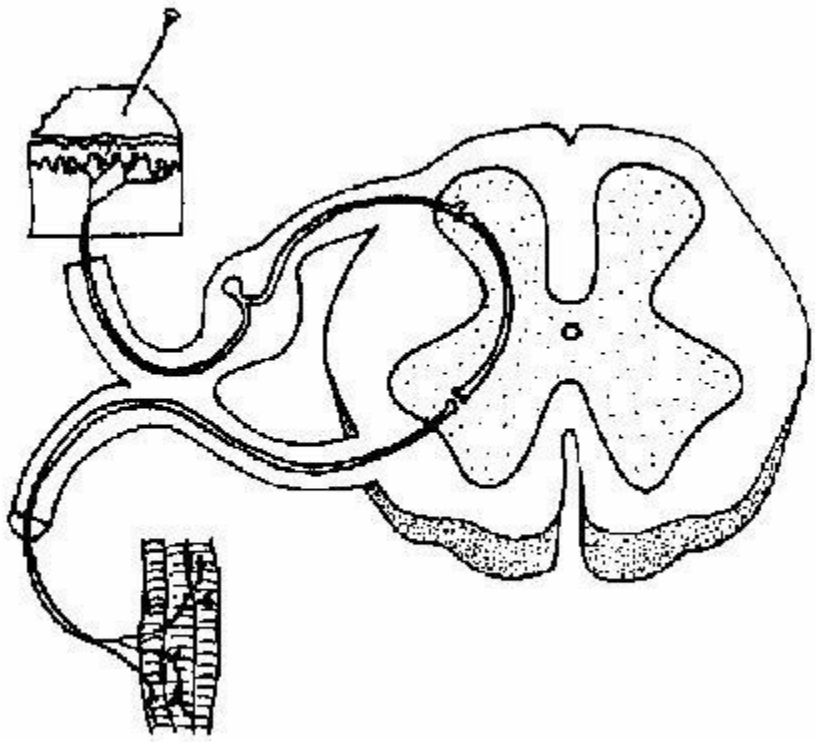
A. Synapse; B. Axon; C. Myelin sheath; D. Nerve impulse; E. Sense receptor; F. Response;
G.Reflex; H. Cell body; I. Dendrite; J. Nerve; K. Neurotransmitter; L. Axon terminal

Term	Function
.....	1. The long fibre that carries the nerve impulses.
.....	2. A bundle of axons.
.....	3. The connection between adjacent neurons.
.....	4. The chemical secreted into the gap between neurons at a synapse.
.....	5. A rapid automatic response to a stimulus.

.....	6. The covering of fatty material that speeds up the passage of nerve impulses.
.....	7. The structure at the end of an axon that produces neurotransmitters to transmit the nerve impulse across the synapse.
.....	8. The high speed signals that pass along the axons of nerve cells.
.....	9. The branching filaments that conduct nerve impulses towards the cell.
.....	10. The sense organ or cells that receive stimuli from within and outside the body.
.....	11. The reaction to a stimulus by a muscle or gland.
.....	12.The part of the nerve cell containing the nucleus.

4. The diagram below shows a cross-section of the spinal cord. Add the following labels to the diagram.

Central canal; White matter; Dorsal root; Grey matter; Ventral root; Skin; Muscle; Sensory neuron; Relay neuron; Motor neuron; Pain receptors in skin



5.

a) List in order the 3 different neurons involved in a reflex arc from the stimulus to the response.

Stimulus	.....	.....	.....	**Response**